



ULTRA COLLEGE OF ENGINEERING AND TECHNOLOGY

Department Of Computer Science Engineering

Completed a project on

**Optimizing User, Group, and Role Management with
Access Control and Workflows**

Submitted By

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Optimizing User, Group, and Role Management with Access Control and Workflows

Problem Statement:

In a small project management team consisting of a Project Manager (Alice) and a Team Member (Bob), there is a need to efficiently manage project tasks and ensure accountability throughout the project lifecycle. The current system lacks clear role definitions, access controls, and a structured workflow, leading to confusion regarding task assignments and progress tracking.

Create Users

1. Open service now
2. Click on All >> search for users
3. Select Users under system security
4. Click on new
5. Fill the following details to create a new user
6. Click on submit

ServiceNow User Administration interface for user 'alice p'. The User ID field is highlighted with a red box. The form includes fields for First name, Last name, Title, Department, Email, Language, Calendar integration, Time zone, Date format, Business phone, and Mobile phone. There are also checkboxes for Password needs reset, Locked out, Active, Web service access only, and Internal Integration User. Buttons for Update, Set Password, and Delete are visible.

7.Create one more user:

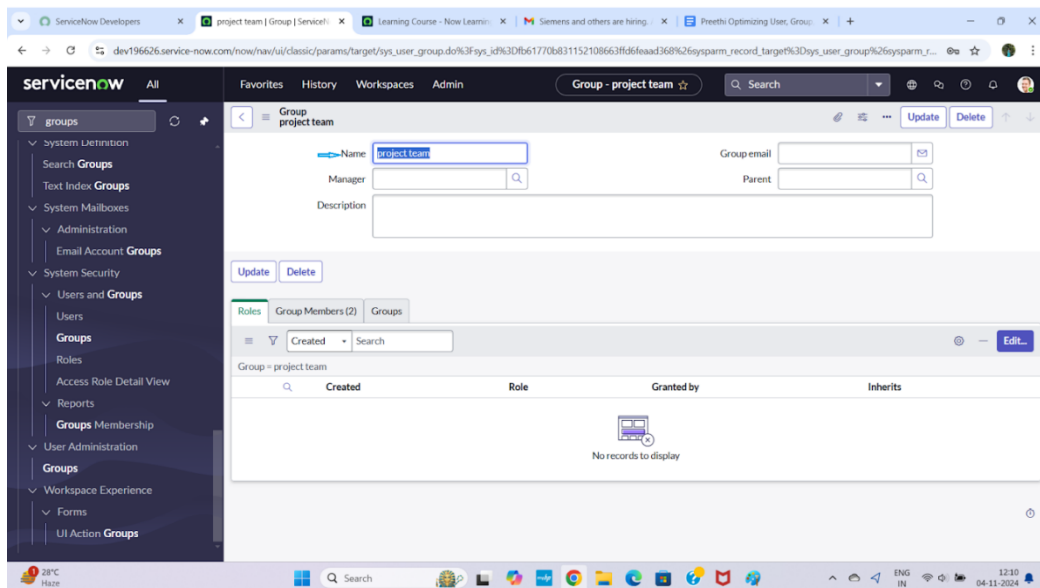
8.Create another user with the following details

9.Click on submit

ServiceNow User Administration interface for user 'Bob p'. The User ID field is highlighted with a red box. The form includes fields for First name, Last name, Title, Department, Email, Language, Calendar integration, Time zone, Date format, Business phone, and Mobile phone. There are also checkboxes for Password needs reset, Locked out, Active, Web service access only, and Internal Integration User. Buttons for Update, Set Password, and Delete are visible.

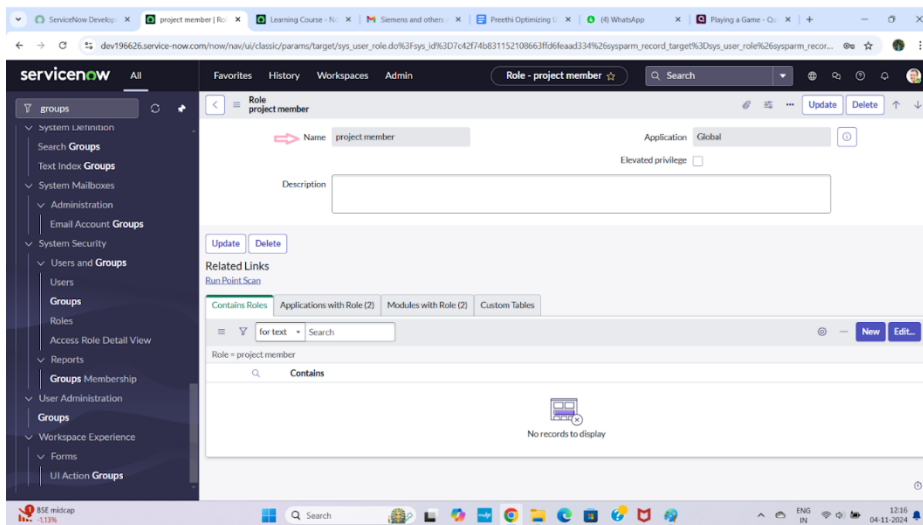
Create Groups

1. Open service now.
2. Click on All >> search for groups
3. Select groups under system security
4. Click on new
5. Fill the following details to create a new group
6. Click on submit



Create Roles

1. Open service now.
2. Click on All >> search for roles
3. Select roles under system security
4. Click on new
5. Fill the following details to create a new role
6. Click on submit



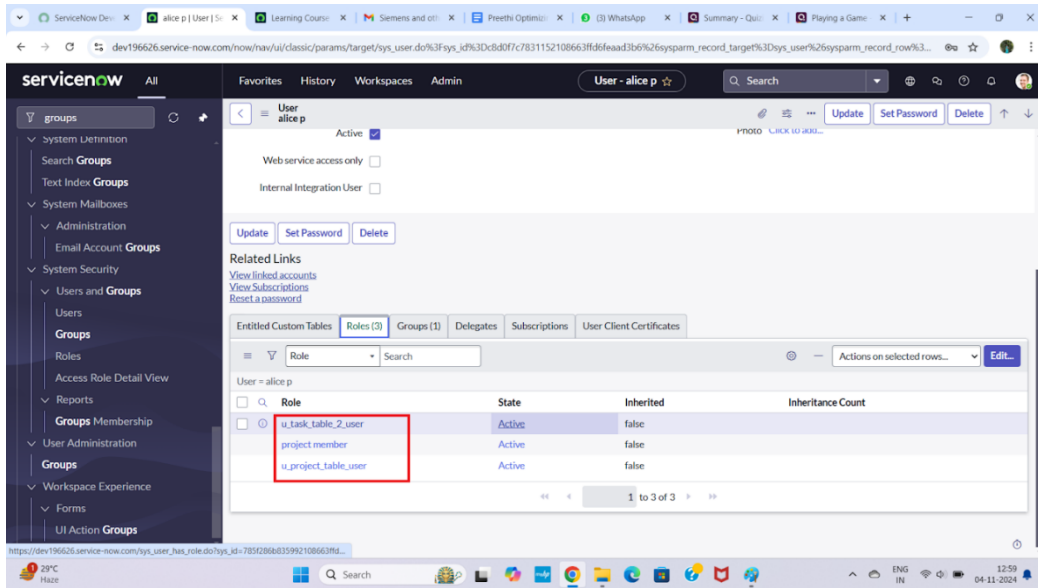
Create one more role:

7. Create another role with the following details

8. Click on submit

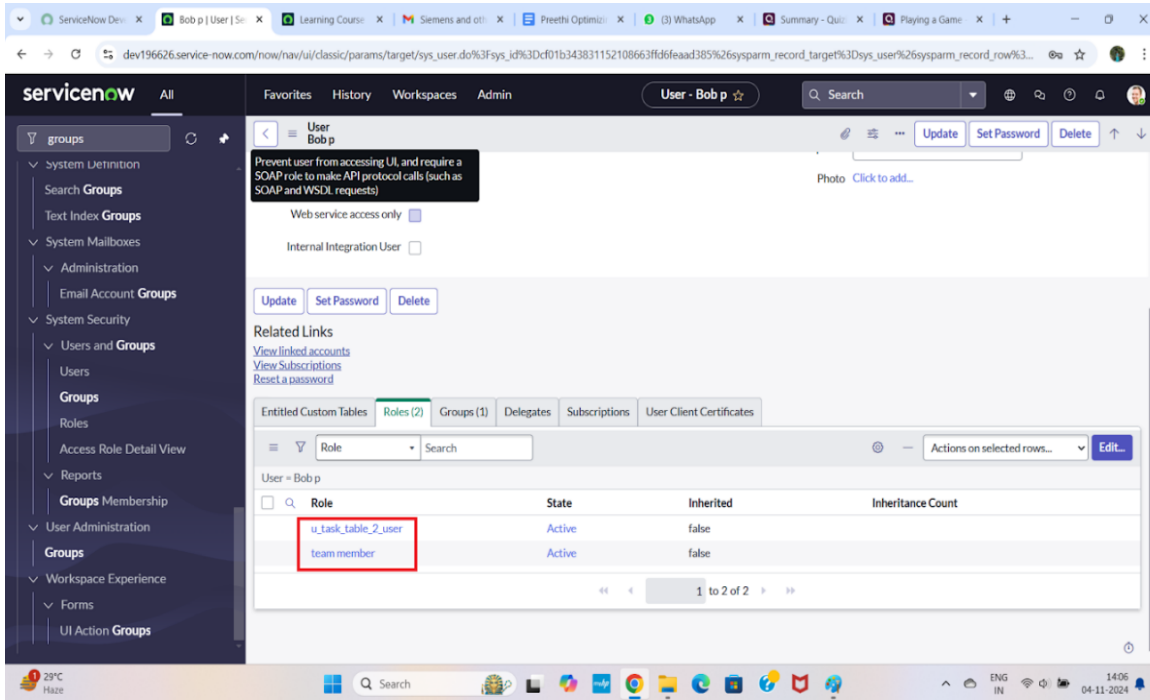
Assign roles to alice user

1. Open servicenow. Click on All >> search for user
2. Select tables under system definition
3. Select the project manager user
4. Under project manager
5. Click on edit
6. Select project member and save
7. click on edit add u_project_table role and u_task_table role
8. click on save and update the form.



Assign roles to bob user

1. Open servicenow. Click on All >> search for user
2. Select tables under system definition
3. Select the bob p user
4. Under team member
5. Click on edit
6. Select team member and give table role and save
7. Click on profile icon Impersonate user to bob
8. We can see the task table2.



Assign table access to application

1. while creating a table it automatically create a application and module for that table
2. Go to application navigator search for search project table application
3. Click on edit module
4. Give project member roles to that application
5. Search for task table2 and click on edit application.
6. Give the project member and team member role for task table 2 application

dev196626.service-now.com/now/nav/ui/classic/params/target/sys_app_application.do%3Fsys_id%3D9705334f831152108663ffd6feaad362

servicenow All Favorites History Admin Application Menu - project table Search

Application Menu project table Update Delete

An application menu is a group of modules in the application navigator. Choose the roles that are required to access the application and add or remove modules in the related list below. [More Info](#)

* Title Application Active ☒

Restricts access to the specified roles. Otherwise, all users can view the application menu when it is active.

Roles project member

Specifies the [menu category](#), which defines the navigation menu style. The default value is Custom Applications.

Category

The text that appears in a tooltip when a user points to this application menu

Hint

Description

Update Delete

Activate Windows
Go to Settings to activate Windows.

dev196626.service-now.com/now/nav/ui/classic/params/target/sys_app_application.do%3Fsys_id%3D114bece3835992108663ffd6feaad3dc

servicenow All Favorites History Admin Application Menu - task table 2 Search

Application Menu task table 2 Update Delete

* Title Application Active ☒

Restricts access to the specified roles. Otherwise, all users can view the application menu when it is active.

Roles u_task_table_2_user, project member, team member

Specifies the [menu category](#), which defines the navigation menu style. The default value is Custom Applications.

Category

The text that appears in a tooltip when a user points to this application menu

Hint

Description

Update Delete

Modules Order Search

Activate Windows
Go to Settings to activate Windows.

Actions on selected rows... New

Create ACL

1. Open service now.
2. Click on All >> search for ACL
3. Select Access Control(ACL) under system security
4. Click on elevate role
5. Click on new
6. Fill the following details to create a new ACL

The screenshot shows the 'Access Control - New Record' form in ServiceNow. The form is titled 'Access Control - New Record' and has a 'Submit' button. A warning message at the top states: 'Warning: A role, security attribute, data condition, or script is required to properly secure access with this ACL.' The form fields are as follows:

- * Type: record
- * Operation: write
- Decision Type: Allow If
- Application: Global
- Active: ☒
- Advanced: ☐
- Admin overrides: ☒
- Protection policy: -- None --
- * Name: task table 2 [u_task_table_2]
- status: dropdown menu
- Description: text area
- Applies To: No. of records matching the condition: 1
- Buttons: Add Filter Condition, Add "OR" Clause
- Footer: Conditions, Activate Windows, Go to Settings to activate Windows.

7. Scroll down under requires role
8. Double click on insert a new row
9. Give task table and team member role
10. Click on submit
11. Similarly create 4 acl for the following fields

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_leave_request	Allow If	delete	record	true	admin	2024-10-22 02:27:59
u_leave_request	Allow If	create	record	true	admin	2024-10-22 02:27:59
u_task_table	Allow If	read	record	true	admin	2024-10-22 04:21:28
u_task_table	Allow If	write	record	true	admin	2024-10-22 04:20:15
u_task_table.u_assigned_to	Allow If	write	record	true	admin	2024-10-22 04:33:53
u_task_table.u_due_date	Allow If	write	record	true	admin	2024-10-22 04:33:14
u_task_table.u_task_id	Allow If	write	record	true	admin	2024-10-22 04:27:47
u_task_table.u_task_name	Allow If	write	record	true	admin	2024-10-22 04:31:14
u_task_table.2	Allow If	write	record	true	admin	2024-10-22 21:05:07
u_task_table.2	Allow If	read	record	true	admin	2024-10-22 21:26:57
u_task_table.2	Allow If	read	record	true	admin	2024-10-22 21:05:07
u_task_table.2	Allow If	write	record	true	admin	2024-10-22 21:28:27
u_task_table.2	Allow If	create	record	true	admin	2024-10-22 21:05:06
u_task_table.2	Allow If	delete	record	true	admin	2024-10-22 21:05:07
u_task_table.2.u_assigned_to	Allow If	write	record	true	admin	2024-10-22 21:31:20

12. Click on profile on top right side

13. Click on impersonate user

14. Select bob user

15. Go to all and select task table2 in the application menu bar

16. Comment and status fields are have the edit access

task table 2 - Create Created

task id:

task name:

status:

assigned to:

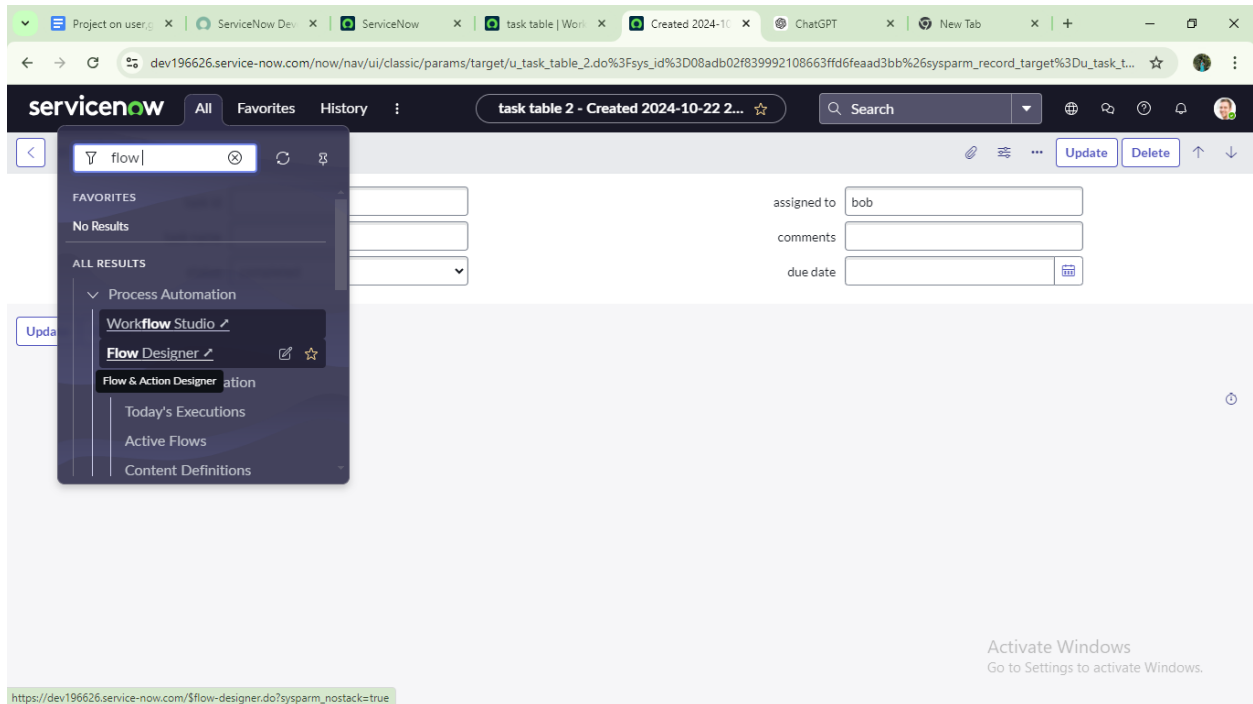
comments:

due date:

Submit

Create a Flow to Assign operations ticket to group

1. Open service now.
2. Click on All >> search for Flow Designer
3. Click on Flow Designer under Process Automation.
4. After opening Flow Designer Click on new and select Flow.
5. Under Flow properties Give Flow Name as “ task table”.
6. Application should be Global.
7. Click build flow.



Project on user...ServiceNow Dev...ServiceNow...Homepage - Flow...Created 2024-1...ChatGPT...New Tab...+

dev196626.service-now.com/now/workflow-studio/home/flow

Workflow Studiotask tableFlow

HomepageOperationsIntegrations

PlaybooksFlowsSubflowsActionsDecision tables

Flows 39Last refreshed just now

New

Playbook

Flow

Subflow

Action

Decision table

Pick up where you left off

Name	Application	Status	Active	Updated
Benchmark Recommendation Evaluator	Benchmarks Spoke	Published	true	2024-09-27 22:00:19
Business process approval flow	Global	Published	true	2020-09-27 22:00:19
Change - Cloud Infrastructure - Authorize	Global	Published	true	2020-11-11 07:08:05
Change - Emergency - Authorize	Global	Published	true	2020-10-06 05:39:49
Change - Emergency - Implement	Global	Published	true	2020-09-23 05:06:26
Change - Emergency - Review	Global	Published	true	2020-10-27 04:18:08
Change - Normal - Assess	Global	Published	true	2020-10-06 05:37:05
Change - Normal - Authorize	Global	Published	true	2020-10-06 05:38:35
Change - Normal - Implement	Global	Published	true	2020-09-23 04:23:59

task table

Last updated: 14 min. ago by Syst...

Create Flow Data

Last updated: 5 months ago by Sy...

Steps

Last updated: 5 months ago by Sy...

Latest updates

System Administrator modified task table

14 min. ago

System Administrator modified Create Flow Data

5 months ago

System Administrator modified Steps

Settings to activate Windows.

5 months ago

System Administrator modified

Project on user...ServiceNow Dev...ServiceNow...New Flow | Workflow Studio...Created 2024-1...ChatGPT...New Tab...+

dev196626.service-now.com/now/workflow-studio/builder?typeSysId=2d85e527439231106c4bb0117fb8f208&sysId=-1

Workflow Studiotask tableFlowOperationsNew Flow

+

+

Let's get the details for your flow

Flow name *

task table

Description

Describe your flow.

Application *

Global

Show additional properties

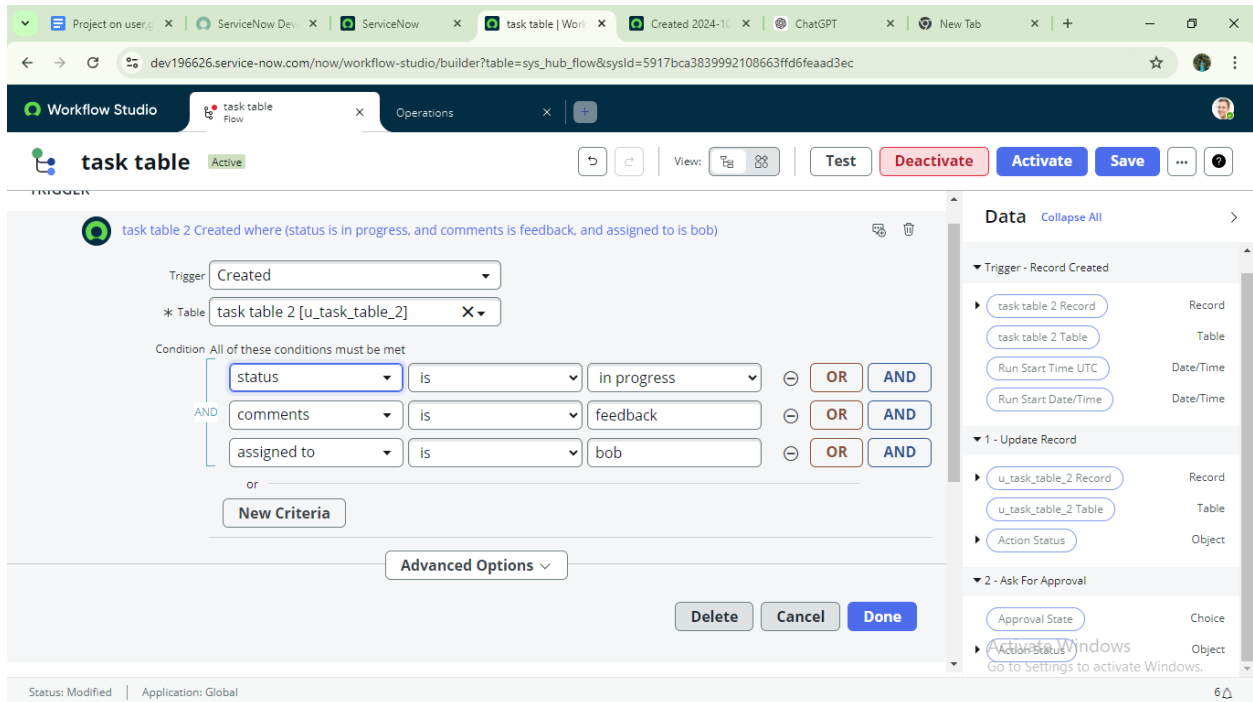
Activate Windows

Go to Settings to activate Windows

CancelBuild flow

next step:

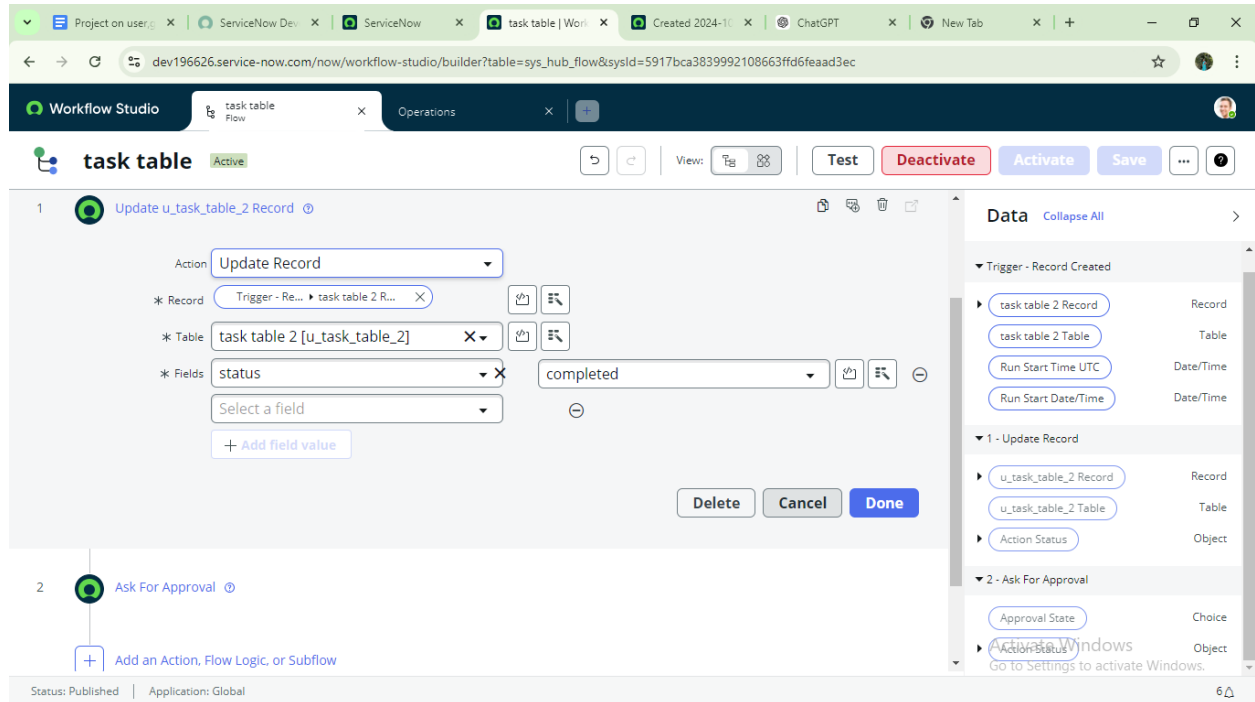
1. Click on Add a trigger
2. Select the trigger in that Search for “create record” and select that.
3. Give the table name as “ task table ”.
4. Give the Condition as Field : status Operator :is Value : in progress
Field : comments Operator :is Value : feedback
Field : assigned to Operator :is Value : bob
5. After that click on Done.



Next step:

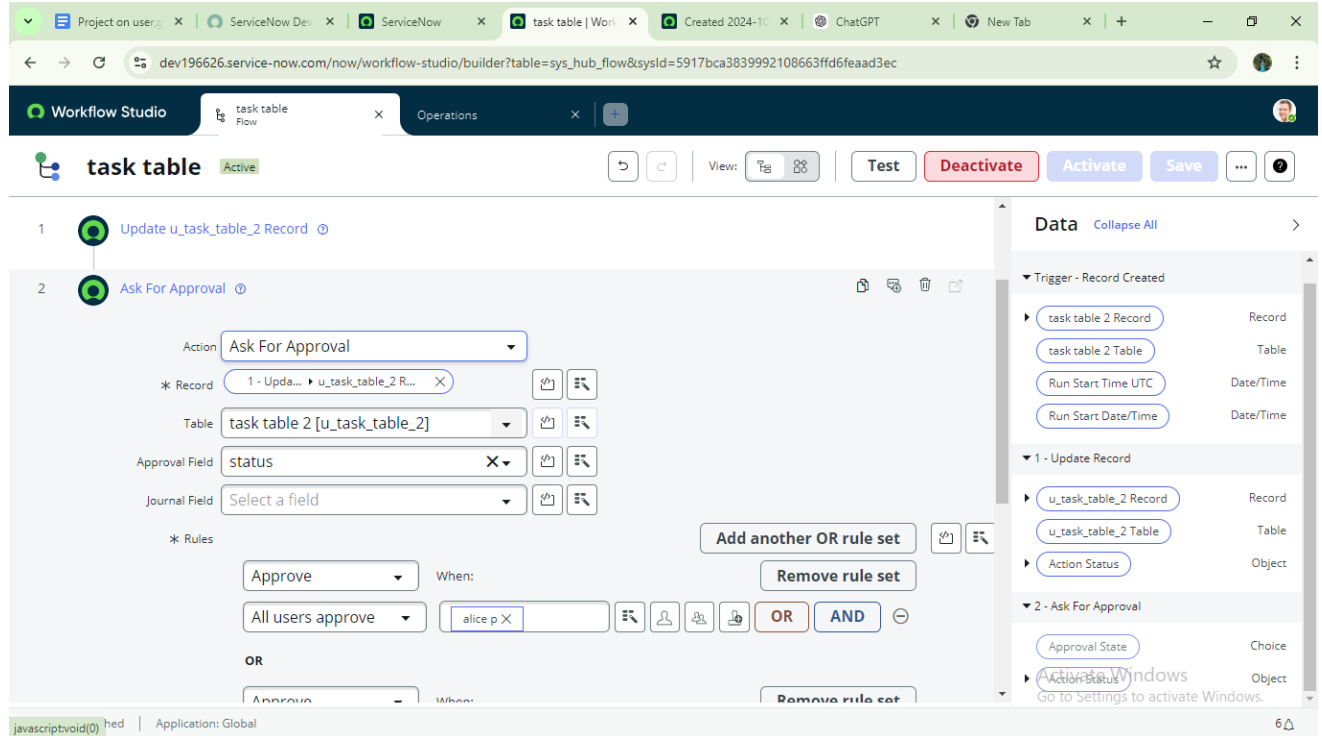
1. Click on Add an action.
2. Select action in that ,search for “ update records”.
3. In Record field drag the fields from the data navigation from Right Side(Data pill)
4. Table will be auto assigned after that

5. Add fields as “status” and value as “completed”
6. Click on Done.



Next step:

1. Now under Actions.
2. Click on Add an action.
3. Select action in that ,search for “ ask for approval ”.
4. In Record field drag the fields from the data navigation from Right side
5. Table will be auto assigned after that
6. Give the approve field as “ status”
7. Give approver as alice p
8. Click on Done.



1. Go to application navigator search for task table.
2. It status field is updated to completed

task id

task name

status

assigned to

comments

due date

Update Delete

Update Delete

1. Go to application navigator and search for my approval
2. Click on my approval under the service desk.
3. Alice p got approval request then right click on requested then select approved

	State	Approver	Comments	Approval for	Created
☐	Approved	alice p		(empty)	2024-10-22 22:26:19
☐	Rejected	Fred Luddy		(empty)	2024-09-01 12:19:33
☐	Requested	Fred Luddy		(empty)	2024-09-01 12:17:03
☐	Requested	Fred Luddy		(empty)	2024-09-01 12:15:44
☐	Requested	Howard Johnson		CHG0000096	2024-09-01 06:15:29
☐	Requested	Ron Kettering		CHG0000096	2024-09-01 06:15:29
☐	Requested	Luke Wilson		CHG0000096	2024-09-01 06:15:29
☐	Requested	Christen Mitchell		CHG0000096	2024-09-01 06:15:29
☐	Requested	Bernard Laboy		CHG0000096	2024-09-01 06:15:29
☐	Requested	Howard Johnson		CHG0000095	2024-09-01 06:15:25
☐	Requested	Ron Kettering		CHG0000095	2024-09-01 06:15:25
☐	Requested	Luke Wilson		CHG0000095	2024-09-01 06:15:25
☐	Requested	Christen Mitchell		CHG0000095	2024-09-01 06:15:25
☐	Requested	Bernard Laboy		CHG0000095	2024-09-01 06:15:25

Conclusion :

This scenario highlights a structured approach to project management, showcasing the roles of Alice and Bob within a defined workflow. With Alice's oversight and Bob's execution, the team effectively collaborates to ensure project success. The use of tables organizes key information, facilitating easy tracking of projects, tasks, and progress updates. Overall, this system promotes accountability, enhances communication, and leads to the successful completion of projects.